Chapter 1

Combinatorial analysis

1.1 Introduction

Combinatorial analysis aims to count the different arrangements that can be formed from a set of elements. It is the branch of mathematics that studies how to **count** objects **without listing them one by one**. When there are five possibilities, we can write them all down; when there are ten thousand or a million, we need formulas. This chapter provides those formulas.

Why count in a probability course? When all the outcomes of an experiment are equally likely, the probability of an event A is computed as a ratio:

$$P(A) = \frac{\text{number of favourable outcomes}}{\text{number of possible outcomes}} = \frac{\text{Card}(A)}{\text{Card}(\Omega)}. \tag{1.1}$$

Both the numerator and the denominator are counts. Without counting, there is no probability computation: combinatorial analysis is the basic tool of the whole course.

Example 1.1 *(Licence plates)*
The licence plates of a certain country consist of three letters followed by three digits. How many different plates can be made?

Writing them all down is out of the question. We will answer this question in Section 1.3, using the multiplication principle.

1.2 Basic notions

Definition 1.1 (Factorial)

The factorial of a non-negative integer n is the product of all positive integers less than or equal to n:

$$n! = n \times (n-1) \times (n-2) \times \cdots \times 2 \times 1. \tag{1.2}$$

By convention, the factorial of 0 equals 1, that is $0! = 1$.

Example 1.2
The first values of the factorial are:

n	0	1	2	3	4	5	6	10
$n!$	1	1	2	6	24	120	720	3 628 800

The factorial grows extremely fast: 10! already exceeds three million.

Remark 1.1 (Simplify before computing)

For every integer $n \geq 1$, we have $n! = n \times (n-1)!$. In a ratio of factorials, we **never** compute the factorials in full: we simplify first. For example,

$$\frac{10!}{8!} = \frac{10 \times 9 \times 8!}{8!} = 10 \times 9 = 90.$$

Definition 1.2 (Set)

A set is a collection of objects called elements.

Definition 1.3 (Subset)

F is a subset of the set E (or F is included in E) if every element of F is also an element of E. This is written $F \subset E$.

Definition 1.4 (Cardinality)

The number of elements of a finite set E is called the cardinality of E, written $\mathrm{Card}(E)$ or $|E|$. The empty set, written $\emptyset$, is the set that contains no element: $\mathrm{Card}(\emptyset) = 0$.

Example 1.3
Let $A = \{1, 2, 3, a, b, c\}$ and $B = \{1, 2, 3\}$. Every element of B belongs to A, so $B \subset A$. Moreover, $\mathrm{Card}(A) = 6$ and $\mathrm{Card}(B) = 3$.

Throughout this chapter, **counting** means computing the cardinality of a finite set.

1.3 Fundamental counting principles

Two simple principles are enough to solve most counting problems. All the formulas of the following sections follow from them.

1.3.1 Addition principle

We use the addition principle when we make a single choice, but this choice may come from several categories that have no element in common. The keyword to look for in a problem is "or".

Proposition 1.1 (Addition principle)

If a choice can be made **either** in m ways from a first category, **or** in n ways from a second category, the two categories being disjoint, then there are $m+n$ possible choices. In other words, if $A \cap B = \emptyset$, then

$$\mathrm{Card}(A \cup B) = \mathrm{Card}(A) + \mathrm{Card}(B). \tag{1.3}$$

Example 1.4 *(At the university cafeteria)*
The university cafeteria offers 3 meat dishes and 4 fish dishes. A student chooses **a single** dish. He takes a meat dish **or** a fish dish: he therefore has $3 + 4 = 7$ possible choices.

1.3.2 Multiplication principle

The multiplication principle, also called the fundamental counting principle, counts the outcomes of an experiment that splits into a sequence of sub-experiments. The keywords are "and", "then".

Proposition 1.2 (The case of two experiments)

Suppose two experiments are to be performed. If experiment 1 can produce any one of m outcomes and if, for each of them, there are n possible outcomes for experiment 2, then there are $m \times n$ outcomes for the two experiments taken together.

Example 1.5 *(Outfits)*
A student owns 3 shirts S_1, S_2, S_3 and 2 pairs of trousers T_1, T_2. He chooses a shirt **then** a pair of trousers: he can wear $3 \times 2 = 6$ different outfits. The choice tree in Figure 1.1 explains why we multiply: from each of the 3 shirts come 2 branches, one per pair of trousers, which gives $3 \times 2 = 6$ endpoints.

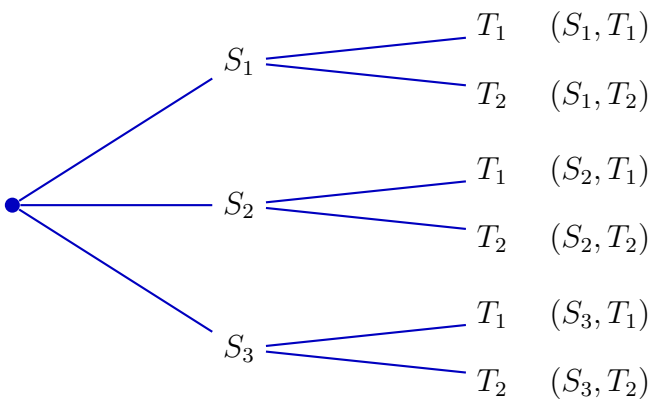


Figure 1.1: Choice tree: 3 shirts, then 2 pairs of trousers

When there are more than two experiments, the principle generalises as follows.

Proposition 1.3 (The case of m experiments)

Suppose an experiment is a sequence of m sub-experiments. If the i-th sub-experiment has n_i possible outcomes, for $i = 1, \ldots, m$, then the total number N of possible outcomes of the whole experiment is

$$N = \prod_{i=1}^{m} n_i = n_1 \times n_2 \times \cdots \times n_m. \tag{1.4}$$

Example 1.6 *(The padlock)*
A padlock has 3 dials, each showing the digits 0 to 9. We make three successive choices:

- first digit of the code (1st sub-experiment): $n_1 = 10$ possibilities;
- second digit of the code (2nd sub-experiment): $n_2 = 10$ possibilities;
- third digit of the code (3rd sub-experiment): $n_3 = 10$ possibilities.

The number of different codes is the product of the possibilities of each sub-experiment:

$$N = n_1 \times n_2 \times n_3 = 10 \times 10 \times 10 = 1\,000.$$

Example 1.7 *(Back to the licence plates)*
Let us return to Example 1.1. A plate is built by six successive choices: three letters (26 possibilities each), then three digits (10 possibilities each). By (1.4):

$$N = 26 \times 26 \times 26 \times 10 \times 10 \times 10 = 26^3 \times 10^3 = 17\,576\,000 \text{ plates.}$$

If we now require **two distinct** letters followed by three digits whose first digit is **not zero**, each choice restricts the following ones: 26 choices for the first letter, 25 for the second, 9 for the first digit, then 10 and 10. We get $26 \times 25 \times 9 \times 10 \times 10 = 585\,000$ plates.

Remark 1.2 (Add or multiply?)

"Or" (a single choice among disjoint cases) $\Longrightarrow$ we **add**.
"And", "then" (successive choices) $\Longrightarrow$ we **multiply**.
Most exercises combine these two rules.

1.4 How to choose the right formula

From now on, we choose p elements among the n elements of a set E. Before applying a formula, we ask two questions:

1. **Does the order matter?** For a code, yes: 12 and 21 are two different codes. For a team, no: {Amine, Yacine} and {Yacine, Amine} are the same team. If the order matters, we speak of an arrangement (or a permutation when $p = n$); otherwise, of a combination.
2. **Can an element be chosen several times?** In a code, a digit may be repeated: this is a choice with repetition. A person cannot occupy two places on a podium: this is a choice without repetition.

The answers to these two questions lead to the right formula (Figure 1.2). The formulas themselves are established in the following sections.

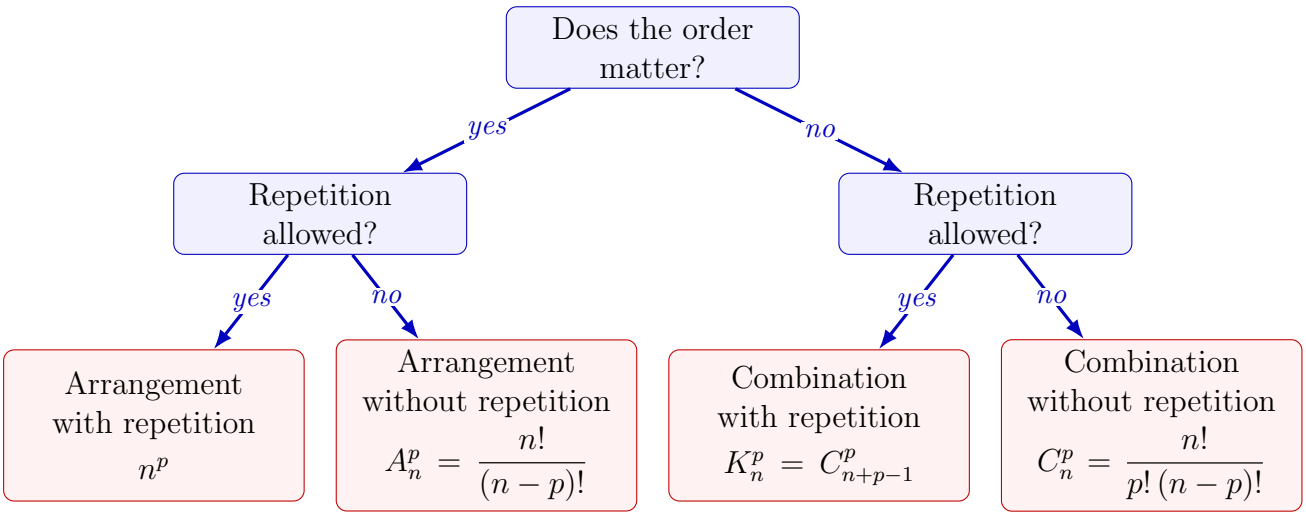


Figure 1.2: Two questions to choose the counting formula

Remark 1.3 (Drawing balls from an urn)

Problems often describe the situation as drawing p balls from an urn containing n. You must know how to translate:

Type of draw	**Order**	**Repetition**	**Formula**
Successive **with** replacement	matters	yes	n^p
Successive **without** replacement	matters	no	A_n^p
Simultaneous (all at once)	does not matter	no	C_n^p

1.5 Counting formulas

1.5.1 Arrangements

Definition 1.5 (Arrangement)

Let E be a finite set with n elements and p a non-negative integer. An arrangement of p elements of E is any ordered sequence of p elements taken from the n elements of E.

There are two cases, depending on whether the same element may appear several times in the sequence or not.

1.5.1.1 Arrangements with repetition

Theorem 1.1 (Arrangements with repetition)

When an element may appear several times in the sequence, the number of arrangements with repetition of p elements out of n is

$$\widetilde{A}_n^p = n^p. \tag{1.5}$$

Proof. We fill the p positions of the sequence one after the other. Since repetition is allowed, each position can receive any of the n elements. By the multiplication principle, the number of sequences is $n \times n \times \cdots \times n = n^p$. □

Example 1.8 *(Passwords)*
We want to build a password made of 3 digits. The order of the digits matters: the password 112 is different from the password 211. Moreover, a digit may be repeated. The number of passwords is therefore the number of arrangements with repetition, with $p = 3$ and $n = 10$:

$$\widetilde{A}_{10}^3 = 10^3 = 1\,000.$$

Example 1.9 *(Guessing on a multiple-choice test)*
A multiple-choice test has 4 questions; each question offers 5 answers, of which the student must tick exactly one. An answer sheet is an ordered sequence of 4 answers (one per question), and the same letter may come up again. There are therefore $5^4 = 625$ possible answer sheets.

1.5.1.2 Arrangements without repetition

Theorem 1.2 (Arrangements without repetition)

When each element can appear only once in the sequence (so $0 \leq p \leq n$), the number of arrangements without repetition of p elements out of n is

$$A_n^p = \frac{n!}{(n-p)!} = \underbrace{n \times (n-1) \times \cdots \times (n-p+1)}_{p \text{ factors}}. \tag{1.6}$$

Proof. Again we fill the p positions one after the other. The first position can receive n elements; an element already placed is no longer available, so the second position has only $n-1$ left, the third $n-2$, and the p-th has $n-p+1$. By the multiplication principle,

$$A_n^p = n(n-1)\cdots(n-p+1) = \frac{n(n-1)\cdots(n-p+1) \times (n-p)!}{(n-p)!} = \frac{n!}{(n-p)!}.$$ □

Example 1.10 *(Passwords with distinct digits)*
Let us return to Example 1.8, but now require the password to contain 3 **different** digits. These are arrangements without repetition, with $p = 3$ and $n = 10$:

$$A_{10}^{3} = \frac{10!}{7!} = 10 \times 9 \times 8 = 720.$$

Forbidding repetition removes $1\,000 - 720 = 280$ passwords.

Example 1.11 *(The podium)*
A race has 8 runners. In how many ways can the gold, silver and bronze medals be awarded? The order matters (finishing first is not finishing third) and a runner receives only one medal:

$A_8^3 = 8 \times 7 \times 6 = 336$ possible podiums.

Remark 1.4

In practice, we compute A_n^p with the expanded form of (1.6): we write p decreasing factors starting from n. In the special case $p = n$, we arrange **all** the elements and $A_n^n = \frac{n!}{0!} = n!$. This is one of the reasons for the convention $0! = 1$.

1.5.2 Permutations

Definition 1.6 (Permutation)

Given a set E of n objects, a permutation of these n objects is any ordered sequence of **all** n objects, in other words any arrangement of the n objects taken n at a time.

Again there are two cases, depending on whether all the objects are distinct or not.

1.5.2.1 Permutations without repetition

Theorem 1.3 (Permutations without repetition)

When the n objects are all distinct, the number of permutations is

$$P_n = A_n^n = n!. \tag{1.7}$$

Example 1.12

1. 5 distinct books can be placed on a shelf in $P_5 = 5! = 120$ ways.
2. With the digits $1, 2, 3, 4, 5$, each used exactly once, we can form $P_5 = 5! = 120$ five-digit numbers.

1.5.2.2 Permutations with repetition

What happens if some objects are identical? Swapping two identical objects does not produce a new arrangement. The number $n!$ therefore counts some arrangements several times, and it must be corrected.

Theorem 1.4 (Permutations with repetition)

Consider n objects divided into k groups: n_1 identical objects of type 1, n_2 identical objects of type 2, …, n_k identical objects of type k, with $n_1 + n_2 + \cdots + n_k = n$. The number of permutations of these n objects is

$$\widetilde{P}_n = \frac{n!}{n_1!\, n_2! \cdots n_k!}. \tag{1.8}$$

Proof. Temporarily number the identical objects to make them all distinct: we get $n!$ arrangements. An arrangement "without numbers" corresponds to exactly $n_1!$ ways of ordering the type-1 objects among themselves, $n_2!$ ways for the type-2 objects, and so on. Each arrangement without numbers is therefore counted $n_1!\, n_2! \cdots n_k!$ times among the $n!$, hence (1.8). □

Example 1.13 *(The books)*
Among 5 books, there are 2 identical mathematics books and 2 identical computer science books; the fifth book is unique. The number of ways to shelve them is

$$\widetilde{P}_5 = \frac{5!}{2!\, 2!\, 1!} = \frac{120}{4} = 30.$$

Example 1.14 *(Anagrams)*

1. The word "ESSES" has 5 letters: E twice and S three times. It has $\dfrac{5!}{2!\, 3!} = \dfrac{120}{2 \times 6} = 10$ anagrams.
2. The word "ESSENCE" has 7 letters: E three times, S twice, N once and C once. It has $\dfrac{7!}{3!\, 2!\, 1!\, 1!} = \dfrac{5\,040}{12} = 420$ anagrams.

Example 1.15 *(The chess tournament)*
Among the 10 players of a chess tournament there are 4 Russians, 3 Americans, 2 English players and 1 Algerian. If the ranking shows only the nationalities, the number of different rankings is

$$\frac{10!}{4!\,3!\,2!\,1!} = 12\,600.$$

1.5.3 Combinations

When **the order does not matter**, several arrangements describe the same choice. Let us compare on a small example.

Example 1.16 *(Arrangement or combination?)*
We choose 2 distinct letters from $\{a, b, c\}$.

- **If the order matters**, we get $ab,\ ba,\ ac,\ ca,\ bc,\ cb$, that is $A_3^2 = 6$ arrangements.
- **If the order does not matter**, ab and ba describe the same set $\{a, b\}$. Only $\{a, b\},\ \{a, c\},\ \{b, c\}$ remain, that is 3 outcomes.

Each unordered choice of 2 letters was counted $2! = 2$ times among the arrangements: $\dfrac{6}{2} = 3$.

Definition 1.7 (Combination)

Given a set E of n elements and an integer p with $0 \le p \le n$, a combination of p elements of E is any subset of p elements of E: the elements are taken **without order** and **without repetition**.

Theorem 1.5 (Number of combinations)

The number of combinations of p elements out of n, written C_n^p or $\binom{n}{p}$ (read "n choose p"), is

$$C_n^p = \frac{A_n^p}{p!} = \frac{n!}{p!\,(n-p)!}. \tag{1.9}$$

Proof. Let us generalise Example 1.16. To build an arrangement of p elements out of n, we can first choose the subset of p elements (C_n^p ways), then put these p elements in order ($p!$ ways). By the multiplication principle, $A_n^p = C_n^p \times p!$, hence (1.9). □

Example 1.17 *(The exam)*
A student must answer 3 of the 4 questions of an exam. The questions are distinct and the order in which he answers them does not matter. The number of choices is a combination with $p = 3$ and $n = 4$:

$$C_4^3 = \frac{4!}{3!\,(4-3)!} = 4.$$

This can be seen directly: choosing the 3 questions answered amounts to choosing the single question left out, which gives 4 possibilities.

Example 1.18 *(Teams)*
From a group of 12 students, we form a team of 5. A team has no order:

$$C_{12}^5 = \frac{12!}{5!\,7!} = \frac{12 \times 11 \times 10 \times 9 \times 8}{5 \times 4 \times 3 \times 2 \times 1} = 792 \text{ teams.}$$

To compute C_n^p by hand, write the first p factors of $n!$ in the numerator and $p!$ in the denominator.

Example 1.19 *(The committee)*
We want to form a committee of 3 men and 2 women from 7 men and 5 women. We choose the men **and** the women: we combine the combination formula with the multiplication principle.

$$C_7^3 \times C_5^2 = \frac{7!}{3!\,4!} \times \frac{5!}{2!\,3!} = 35 \times 10 = 350 \text{ committees.}$$

Remark 1.5 (Link with permutations with repetition)

With only **two** types of objects, p identical objects of one type and $n-p$ of the other, formula (1.8) gives

$$\frac{n!}{p!\,(n-p)!} = C_n^p.$$

Indeed, arranging the objects amounts to choosing the p positions occupied by the first type. For "ESSES", placing the 2 letters E amounts to choosing 2 positions out of 5: $C_5^2 = 10$.

1.5.3.1 Properties of combinations

Theorem 1.6 (Properties of combinations)

For all integers $n \geq 1$ and $0 \leq p \leq n$:

1. $C_n^0 = C_n^n = 1$ and $C_n^1 = C_n^{n-1} = n$;
2. $C_n^p = C_n^{n-p}$ (Symmetry)
3. $C_{n-1}^{p-1} + C_{n-1}^p = C_n^p$, for $1 \leq p \leq n-1$ (Pascal's rule)

Proof. 1. There is only one way to take nothing and only one way to take everything; there are n ways to take a single element.

2. Choosing the p elements we **take** amounts to choosing the $n-p$ elements we **leave**. It can also be checked by computation: $C_n^{n-p} = \dfrac{n!}{(n-p)!\,p!} = C_n^p$.

3. Fix an element x of E. The subsets with p elements fall into two disjoint categories: those that **contain** x (we still choose $p-1$ elements among the $n-1$ others, that is C_{n-1}^{p-1} ways) and those that **do not contain** x (we choose the p elements among the $n-1$ others, that is C_{n-1}^p ways). The addition principle gives the formula. □

Example 1.20 *(Using symmetry)*
Computing C_{20}^{18} directly would require 20!. By symmetry:

$$C_{20}^{18} = C_{20}^2 = \frac{20 \times 19}{2 \times 1} = 190.$$

Pascal's rule lets us compute the C_n^p row by row, without factorials: each number is the sum of the number directly above it and its left neighbour. This gives Pascal's triangle (Table 1.1).

$n \backslash p$	0	1	2	3	4	5	6
0	1						
1	1	1					
2	1	2	1				
3	1	3	3	1			
4	1	**4**	**6**	4	1		
5	1	5	**10**	10	5	1	
6	1	6	15	20	15	6	1

Table 1.1: Pascal's triangle: $C_5^2 = C_4^1 + C_4^2 = 4 + 6 = 10$

1.5.3.2 The binomial theorem

The numbers C_n^p are also called binomial coefficients, because of the following result.

Theorem 1.7 (Binomial theorem)

For all real numbers a, b and every integer $n \geq 0$,

$$(a+b)^n = \sum_{k=0}^{n} C_n^k \, a^k \, b^{n-k}. \tag{1.10}$$

Proof. Expand $(a+b)^n = (a+b)(a+b)\cdots(a+b)$ (n factors). Each term of the expansion is obtained by choosing, in each of the n factors, either a or b. The term $a^k b^{n-k}$ appears as many times as there are ways to choose the k factors that provide a, that is C_n^k times. □

Example 1.21
Row $n = 3$ of Pascal's triangle gives the coefficients $1, 3, 3, 1$:

$$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3.$$

Corollary 1.1 (Number of subsets)

Taking $a = b = 1$ in (1.10), we get

$$\sum_{k=0}^{n} C_n^k = 2^n. \tag{1.11}$$

A set with n elements therefore has 2^n subsets (counting $\emptyset$ and the set itself). For example, $\{a, b, c\}$ has $2^3 = 8$ subsets: $\emptyset$, $\{a\}$, $\{b\}$, $\{c\}$, $\{a, b\}$, $\{a, c\}$, $\{b, c\}$, $\{a, b, c\}$.

1.5.3.3 Combinations with repetition

One last case remains: the order does not matter, but the same element may be chosen several times.

Definition 1.8 (Combination with repetition)

A combination with repetition of p elements out of n is a choice of p elements out of n, **regardless of order**, where the same element may be chosen **several times**.

Theorem 1.8 (Combinations with repetition)

The number of combinations with repetition of p elements out of n is

$$K_n^p = C_{n+p-1}^p = \frac{(n+p-1)!}{p!\,(n-1)!}. \tag{1.12}$$

Idea of the proof. Such a choice is completely described by how many times each element is taken. We encode it as a sequence of p stars ($\star$, one per chosen object) and $n-1$ bars ($|$, which separate the n elements). For example, with $n=3$ flavours V, C, S and $p=4$ scoops:

$$\star\star\,|\,\star\,|\star \longleftrightarrow \text{VVCS}, \qquad \star\star\star|\,|\star \longleftrightarrow \text{VVVS}.$$

Each choice corresponds to exactly one sequence and vice versa. A sequence has $n+p-1$ symbols; it is determined by the positions of the p stars, hence C_{n+p-1}^p sequences. □

Example 1.22 *(The ice-cream shop)*
An ice-cream shop offers 3 flavours: vanilla (V), chocolate (C) and strawberry (S). How many 2-scoop cones can be made, given that a flavour may be taken twice and the order of the scoops does not matter?

$$K_3^2 = C_{3+2-1}^2 = C_4^2 = 6 \text{ cones:} \quad VV,\ CC,\ SS,\ VC,\ VS,\ CS.$$

The list confirms the formula.

Example 1.23 *(Integer solutions of an equation)*
How many solutions does the equation $x_1+x_2+x_3=12$ have in **non-negative** integers? A solution amounts to distributing 12 identical units among 3 variables: it is a combination with repetition of $p=12$ elements out of $n=3$. There are therefore

$$K_3^{12} = C_{14}^{12} = C_{14}^2 = \frac{14\times 13}{2} = 91 \text{ solutions.}$$

1.6 Summary

Table 1.2 gathers all the formulas of the chapter. Faced with an exercise, ask the two questions of Section 1.4, and the table gives the formula.

	Without repetition	**With repetition**
Order matters (arrangements)	$A_n^p = \dfrac{n!}{(n-p)!}$ successive draw without replacement	$\widetilde{A}_n^p = n^p$ successive draw with replacement
Order does not matter (combinations)	$C_n^p = \dfrac{n!}{p!\,(n-p)!}$ simultaneous draw	$K_n^p = C_{n+p-1}^p$
All the elements ($p = n$) (permutations)	$P_n = n!$	$\widetilde{P}_n = \dfrac{n!}{n_1!\,n_2!\cdots n_k!}$ identical objects

Table 1.2: Counting formulas

Example 1.24 *(Three ways of drawing)*
An urn contains 8 balls: 5 red and 3 white. We draw 3 balls. The number of possible outcomes depends on how the draw is made:

1. **simultaneous draw** (order does not matter, no repetition): $C_8^3 = \dfrac{8 \times 7 \times 6}{3 \times 2 \times 1} = 56$;
2. **successive draw without replacement** (order matters, no repetition): $A_8^3 = 8 \times 7 \times 6 = 336$;
3. **successive draw with replacement** (order matters, with repetition): $8^3 = 512$.

Same urn, same balls, three different answers: it is the way of drawing that decides the formula.